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RESEARCH ARTICLE

# Automated Pavement Distress Detection Based on Convolutional Neural Network

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**ABSTRACT** Pavement distress detection is crucial in road health assessment and monitoring. However, there are still some challenges in extracting pavement distress based on deep learning: such as insufficient segmentation, extraction errors and discontinuities. In this paper, we propose DARNet, a network for pavement distress extraction. A Distress Aware Attention Module (DAAM) is proposed to solve the problem of discontinuity in distress extraction due to inaccurate recovery of distress pixels during upsampling. Based on the characteristics of distress morphology, a Refinement Extraction Module (REM) is designed to effectively capture horizontal and vertical pavement damage features by fusing high-level and low-level features, which improves the accuracy of the model in extracting details of pavement damage information. Finally, a Weighted Cross-Entropy Loss function (WCEL) is introduced to assign weights according to the distance of the pixel point to the boundary of the distress, which solves the problem that the traditional cross entropy function treats each pixel point equally. We also propose a set of pavement distress datasets LNTU_RDD_GS, and the experimental results show that DARNet can reach 82.68% mIoU and 90.13% F score in the datasets in this paper, 80.63% mIoU and 88.35% F score in the four public datasets.

**INDEX TERMS** Pavement distress detection, convolutional neural network, multiscale feature fusion, attention mechanisms, pavement distress baseline dataset.

## I. INTRODUCTION

With the rapid development of road transportation, roads have received increasing use. In addition to the influence of climatic conditions, road structure and imposed loads, pavement conditions have declined, This potentially jeopardizes vehicle safety in addition to reducing transportation efficiency [1]; additionally, the need for pavement maintenance has received increasing attention. The traditional method of distress detection involves manual reading and manual drawing, which are highly subjective and labour intensive, making it difficult to objectively make a comprehensive diagnosis of road maintenance conditions. How to extract pavement distress efficiently, Intelligently and accurately solving this issue has become an urgent concern.

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Recently, image processing technology has gained widespread usage, and many algorithms based on image segmentation have been developed to extract pavement distress. Examples include threshold segmentation [2], [3], edge detection and other methods [4], [5]. However, these algorithms are generally effective at extracting pavement distress in specific scenes [6]; moreover, actual pavement scenes are complex, and misidentification can occur easily when minor distress is ignored during image processing. To solve this problem, pavement distress detection techniques based on machine learning have been employed by researchers [7], including artificial neural networks [8], support vector machines [9], [10], and randomized structured forests [11]. However, machine learning features can only satisfy distress detection requirements in some specific situations and need to be reset when new distress environments appear, which still cannot meet industrial demand.

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Deep Learning Excellence Leads to Convolutional Neural Networks for various applications [12], [13], [14], [15]. In the field of speech recognition [16], target detection [17] and other areas have achieved better results; the use of convolutional neural networks to automatically identify pavement distress has become the main research direction, and pavement distress detection has also begun to develop in the direction of automation and rapidity [18]. Zhang et al. [19] initially presented the use of deep learning methods for detecting pavement cracks, and the proposed network is more accurate at extracting distress detection than manual methods are. Subsequent researchers have taken several approaches to increase pavement distress detection accuracy, including the use of semantic segmentation networks with higher accuracy and improving the accuracy of the models by optimizing the training parameters of the network models and selecting appropriate loss functions [20], [21], [22]. To better adapt to the needs of the pavement distress detection task, several researchers have started to specifically design network models [23], [24] to capture the characteristics of pavement distress more efficiently.

While the methods mentioned above have yielded positive outcomes, the following challenges remain in the realm of pavement distress detection:

1) The diversity and scale differences of pavement distresses make them difficult to be effectively extracted and detected, and the complexity of the background information of the pavement makes the extraction of distresses susceptible to interference terms such as stains and shadows.

2) Upsampling may blur or eliminate small cracks during the process of pixel recovery, thus leading to missed recognition.

3) Uneven distribution of pavement distress prospects and backgrounds leads to poor detection accuracy.

The main contributions of this paper are as follows:

1) A CFASPP structure was designed to extract distress features using depth separable convolution at different scales, which improved the model's localization and segmentation performance for distress at different scales.

2) A distress aware attention module was designed to dynamically assign weights based on the morphology of the distresses to ensure that important details related to these distresses are not lost during pixel restoration.

3) An efficient refinement module is proposed to exploit the network's higher order and lower order features to capture the details of pavement distress more accurately during the decoding process.

4) we designed a weighted cross entropy loss function that adjusts the impact of model prediction errors on loss in different regions by calculating the distance between pixel positions and boundary within the image.

5) A multiscenario pavement distress dataset, LNTU_RDD_GS, was established for national and provincial arterial roads.

The remaining organization of this paper is as follows: A summary of earlier studies on pavement distress detection is given in Section II. Following that, Section III outlines our modeling approach. In Section IV, we present and analyze the experimental results. Finally, Section V summarizes our work.

## II. RELATED WORKS

### A. DETECTION OF DEEP LEARNING BASED PAVEMENT DISTRESS

As deep learning continues to progress, many classic semantic segmentation models have achieved remarkable success within the domain of image processing. Researchers have also applied these methods to the field of extracting pavement distress. Zhang et al. [25] proposed an improved version of the SegNet network for fine crack extraction by passing the upper layer features layer by layer to highlight the tiny crack features. A boundary detection module is also designed to learn the residuals of labelled images and intermediate feature maps, thereby improving the accuracy of crack boundary location determination. Tao et al. [26] developed an enhanced UNet pavement crack extraction model that introduces structureless convolution and attention mechanisms and integrates a dense null convolution and attention mechanism module between the encoding and decoding parts, which helps to obtain more crack details and global contextual information, thus improving the accuracy of crack localization. Yu et al. [27] proposed a pixel level pavement crack segmentation method based on RUCNet, which uses the residual block in ResNet instead of the UNet encoder convolution layer to extract more accurate crack features. The SCSE attention module is introduced in the decoding section to enhance the important information features and improve the crack segmentation effect. Sun et al. [28] proposed DMANet, a DeepLabv3+ based network, and designed a multiscale attention module to generate an attention mask, which realizes the dynamic allocation of weights between high-level and low-level feature maps. Compared with fixed weighting, this dynamic weighting strategy allocates the weights of different feature maps more intelligently and significantly improves the performance.

However, the generic network model may still have some limitations in pavement distress detection tasks, especially in the detailed extraction of some distresses. Numerous researchers have subsequently developed specialized pavement crack extraction networks. Han et al. [29] constructed CrackWNet, a network containing a roundtrip sampling block layout with jump levels, to improve the ability to perceive information on different scales by combining different receptive fields and to solve the problems of easy interruption of the pavement crack extraction process and misidentification of background noise. Tang et al. [30] constructed a pixel level pavement crack segmentation method EDNet, which significantly reduces the problem of misrecognition due to the imbalance in the number of cracked pixels and non-cracked pixels by using a self-encoder as a decoder and encoding the cracked images into the corresponding feature maps. Zhang et al. [31] constructed a real time pavement

distress segmentation network ESCNet. It proposes ECS modules instead of traditional convolutional layers, which reduce the number of parameters in the model to improve computational efficiency. The model also introduces parallel maximum pooling and convolution operations to accelerate feature processing. Zhang et al. [32] proposed a three branch based pavement distress segmentation network using spatial streams to capture low-level spatial features, context streams to model high-level contextual relationships, and edge detection streams to refine segmentation results and generate additional boundary maps to extract all types of pavement distresses of different sizes. Zhu et al. [33] proposed RHANet, an efficient automatic segmentation network for pavement cracks that integrates residual and incorporating hybrid attention blocks into the codec architecture to enhance high-level feature extraction. The introduction of hybrid attention blocks helps to fuse the high-level and low-level features and helps the model to better focus on the correct channel and crack regions, which significantly improves the feature representation capability of the model.

The Transformer concept is frequently used in many different domains. The transformer model is being investigated in an increasing number of studies to extract features like pavement distress. CrackFormer [34] employs a SegNet style encoder decoder architecture, where the introduced codec block incorporates a self-attention module to extract contextual data across feature channels with the help of a convolutional kernel, while positional embedding is efficiently utilized to capture the wide range of spatial contextual information required for long-distance interactions. Additionally, a scaling attention module is introduced to combine the outputs of the encoder and decoder blocks, suppressing background features and enhancing crack features. Xu et al. [35] proposed the locally enhanced transformer network LETNet, which employs the transformer model to capture remote dependencies while compensating for the lack of low-level and high-level local features with the help of convolutional stemming and local enhancement modules. A skip connection strategy and an efficient upsampling module are utilized to restore detailed crack information. Xiao et al. [36] proposed a hybrid windowed attentional visual network, CrackFormer, which employs a hybrid windowed self-attention algorithm to capture the local feature semantics of cracks under a dense window while acquiring the global feature semantics under a sparse window, thus significantly enhancing the detail and accuracy of the feature semantics. In addition, a weighted multi-head self-attention mechanism is introduced to readjust the contribution weights of different heads, which further enhances the robustness and saliency of feature encoding. Sun et al. [37] Developed the PCDETR, a lightweight detection framework that fuses convolutional and sequential features. This architecture uses a residual network and the Swin-Transformer as two parallel channels, to extract global and local features and optimized the encoder and decoder to achieve an ensemble prediction of both the location and class information of cracks, which reduced network complexity.

### B. ATTENTION MECHANISM

Deep Convolutional Neural Networks (CNNs) have excelled in computer vision, particularly in tasks like image classification. However, as the depth and complexity of networks increase, training and feature learning become more challenging. To overcome these challenges, SENet [38] was created, which enables the network to be able to adaptively learn the weights of each channel to better capture crucial information within input features and enhance CNN performance. However, SENet considers only the channel dimension and does not address the spatial dimension. Roy et al. [39] proposed SCSENet to focus on both channel and spatial information, empowering the network to effectively model objects of different sizes. With the development of convolutional neural networks, network models have typically become increasingly complex [40], [41]. SKNet [42] uses multiscale convolutional kernels to extract features in a way that responds to this complexity by using a weight attention mechanism. This mechanism adapts to the size of the convolutional kernel, dynamically selecting the kernel to emphasize features at various scales, thus aiding the network in efficiently processing information across different scales. DANet [43] dynamically adjusts the network's attention to the input data by introducing a dual attention mechanism to address both spatial and channel features, allowing it to better perceive object locations, feature boundaries, and critical information within the channel to improve the processing of multiscale information and complex scenes. Many existing approaches focus on creating more intricate attention modules to enhance performance, but this inevitably leads to increased model complexity. ECANet [44] effectively reduces the computational burden of models by increasing the importance of channel features, thus achieving better performance and computational efficiency in addressing the problems associated with computational and storage resources in deep CNNs. Triplet attention [45] introduces three different attention mechanisms focusing on spatial, channel and channel combination features to enhance the performance of deep neural networks through the incorporation of rotation operations and residual transformations while controlling the computational overhead to better encode the dependencies between channels and spatial information. Due to the excellent performance of the attention mechanism in deep learning networks, many studies [46], [47] have incorporated the attention mechanism into pavement distress extraction networks.

### C. FEATURE FUSION

Feature fusion is a key concept in the field of deep learning and computer vision that combines different levels of feature information to improve model performance. SSD [48] initially attempted to detect targets on multilevel pyramidal features by fully utilizing multiscale feature maps from different layers in the network feed forward process to detect objects of different sizes. However, since shallow feature maps contain relatively little semantic information, this

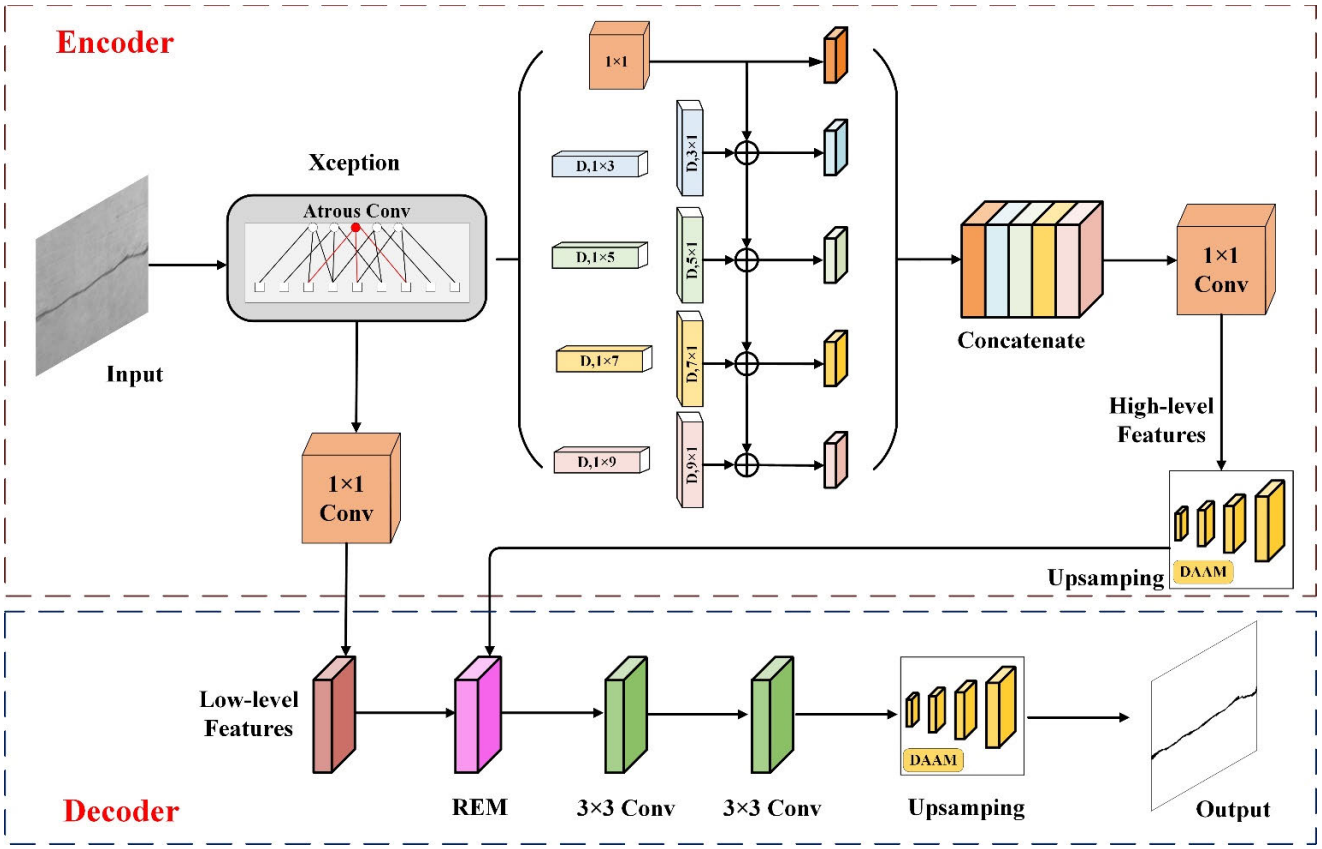



**FIGURE 1.** The structure of DARNet. The four main parts are as follows: (1) a cross feature ASPP Module, (2) two Distress Aware Attention Module (DAAM), (3) a Refinement Extraction Module (REM), and (4) a weighted cross entropy loss function (WCEL).

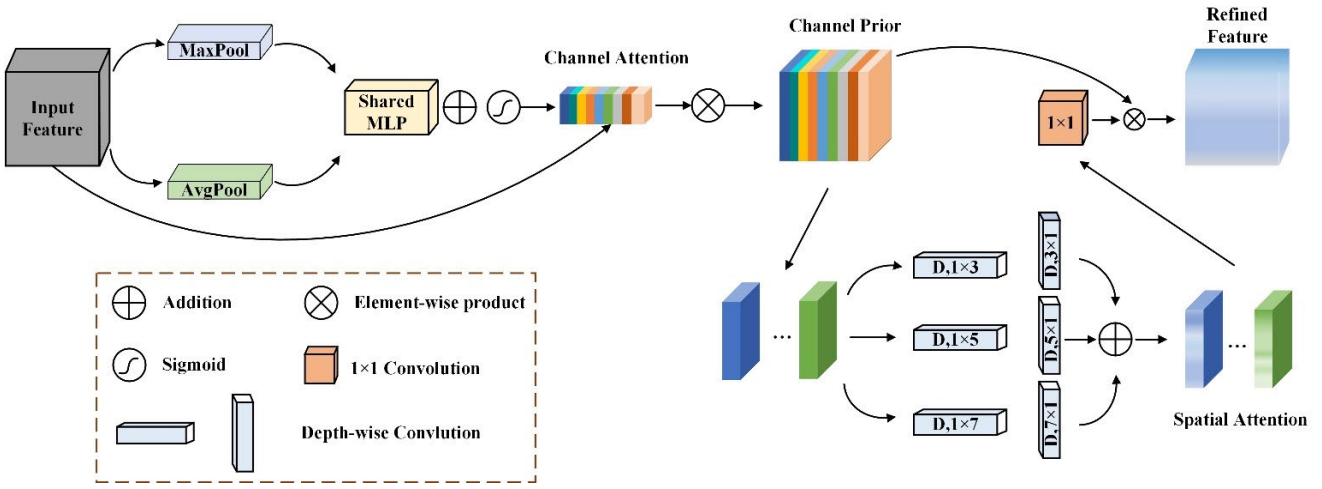



**FIGURE 2.** The structure of DAAM. The weights are dynamically assigned according to the morphology of the defects by using depth-separable convolutions at different scales.

bottom-up approach is less accurate when dealing with small targets. To solve this problem, FPN [49] generates multiple feature maps at different scales and fuses them through top down and bottom-up feature transfer processes, which enables deep learning models to better perceive objects at different scales and improves the performance and efficiency of computer vision applications. The FPN initially introduces a top-down feature map fusion mechanism, but it does not introduce bottom-up fusion. PANet [50] was the first to propose bottom-up feature map fusion while introducing information fusion between lateral and vertical paths. Horizontal paths transfer information across different resolutions,

and vertical paths transfer information between the same resolutions, facilitating multiscale information fusion. The emergence of PANet verified the effectiveness of bidirectional feature fusion, but it used a relatively simple approach, and scholars have attempted more complex bidirectional fusion techniques, such as ASFF [51], NAS-FPN [52] and BiFPN [53]. These methods further improve the performance of multiscale feature fusion for targets of different sizes and complexities. Due to the effectiveness of feature fusion, many works [54], [55], [56] have incorporated it into automatic distress detection, yielding notable progress.

## III. OVERVIEW OF DARNet

This paper build an end to end framework DARNet, that consists of an encoder and a decoder. Xception is used as the backbone of the encoder, and to gain multiscale information, the generated feature maps are loaded into the CFASPP module. Four depth separable convolutions and a normal convolution make up the CFASPP structure. The local information is captured using depth separable convolution, while the broader contextual information is captured using depth separable convolution, and the fusion strategy of neighboring scale feature maps is employed to minimize the loss of feature information, which passes the fused, richly informative features to the next layer for fusion again. Finally, the five feature maps are merged using a cascade operation, and the merged results are subsequently fed into the convolution module to obtain the higher order features. In the decoding stage, the superimposed feature maps are first upsampled and then passed through the refinement extraction module along with the low-order feature output from the encoder's second layer, pass through the refinement extraction module. This process aims to recover the localization of image information through a combination of high-level and low-level feature maps. Finally, the obtained features are convolved and upsampled to obtain the feature maps. Figure 1 illustrates the structure of DARNet.

### A. DISTRESS AWARE ATTENTION MODULE

Upsampling during the process of recovering pixels can not guarantee that each pavement pixel can be completely and accurately recovered to the original position, resulting in the easy loss of small crack like pavement distress at the boundary of the image and thus leading to missed recognition. To address this problem, this paper designs a comprehensive channel and spatial direction of the distress perception attention module. The channel aware attention mechanism assigns different channel weight information so that the model emphasizes the channels that are most important for image reconstruction and distress detection. Orientation aware attention enables the model to concentrate more on the image regions associated with distress morphology, enhances the ability to perceive the spatial structure near the distress, and mitigates the blurring or offset problem that may be introduced during the upsampling process. Figure 2 illustrates the DAAM module.

In this paper, the two attention modules are connected in series, and the image feature $F$ is first derived from the channel weights by the channel attention module. Each channel is assigned a corresponding weight value to the feature $F_c$; then the spatial attention weight is obtained by the spatial attention module. The introduction of the hybrid attention mechanism can make the network more focused on boundary of distress and prevent the loss of pixels when recovering. The expression is as follows:

$$F' = F_c \times F \tag{1}$$

$$F'' = F_{cs} \times F' \tag{2}$$

where $F$ represents the input feature map, $F_c$ is the feature map assigned with channel weights, $F_{cs}$ is the feature map assigned with spatial weights, $F'$ is the feature after the channel attention module, and $F''$ is the feature outputted by the DAAM.

Channel attention allows the model to focus on key pavement distress information in pavement images, thus improving distress characterization. The input image features are subjected to global average pooling and global maximum pooling operations. Then, the features are stretched into a one dimensional vector by a shared MLP, the two processed features are summed, and the activation function sigmoid is used to obtain the channel attention vector $F_c$. The operation process of the channel attention module is shown as follows:

$$F_C = \sigma(\mathrm{MLP}(\mathrm{AvgPool}(F)) + \mathrm{MLP}(\mathrm{MaxPool}(F))) \tag{3}$$

where $F$ is a feature and $\sigma$ employs the sigmoid activation function.

In this paper, spatial attention is proposed based on the morphology of pavement distress. The method employs $3\times3$, $5\times5$ and $7\times7$ depth separable convolutional operations to capture the correlation between positions and the relationship between features, which helps to efficiently address pixel relationships in the image and improves the perception of distress. Moreover, depth separable convolution has fewer parameters than does the ordinary convolution kernel, which improves the computational efficiency. Finally, the model calculates the weighted summation of the weight rows at each scale so that it can automatically learn the weight assignment strategy adapted to different distresses and accurately localize the distresses at different scales; moreover, this approach helps to achieve accurate localization and recovery in terms of pixel restoration. The expression is as follows:

$$F_s = \mathrm{Conv}_{1\times1}(\sum_{i=0}^{3}(\mathrm{DwConv}(F))) \tag{4}$$

where $Dw$ is the depth separable convolution. We visualize the feature map as a heatmap during the training process, which can intuitively show where the model pays more attention. As shown in Figure 3, by using the DAAM module, during training the model pays greater attention to tiny distress locations, demonstrating the effectiveness of the method.

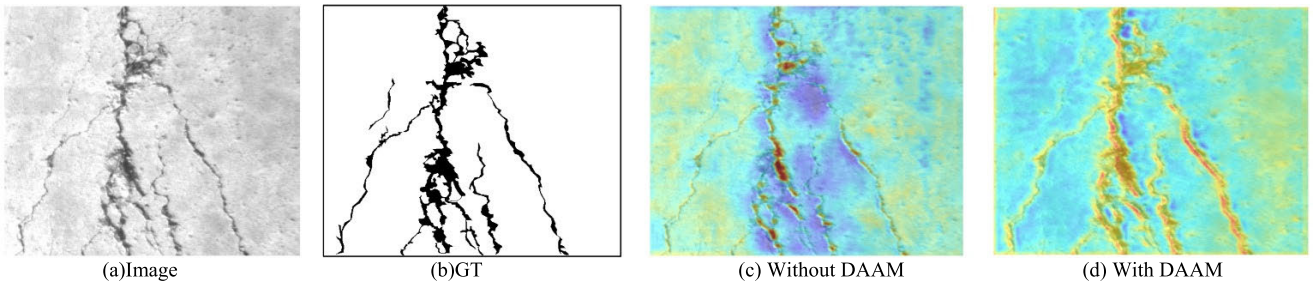



**FIGURE 3.** Feature heatmap visualization during training with or without DAAM addition. It enables the model to focus more on tiny distress in the image.

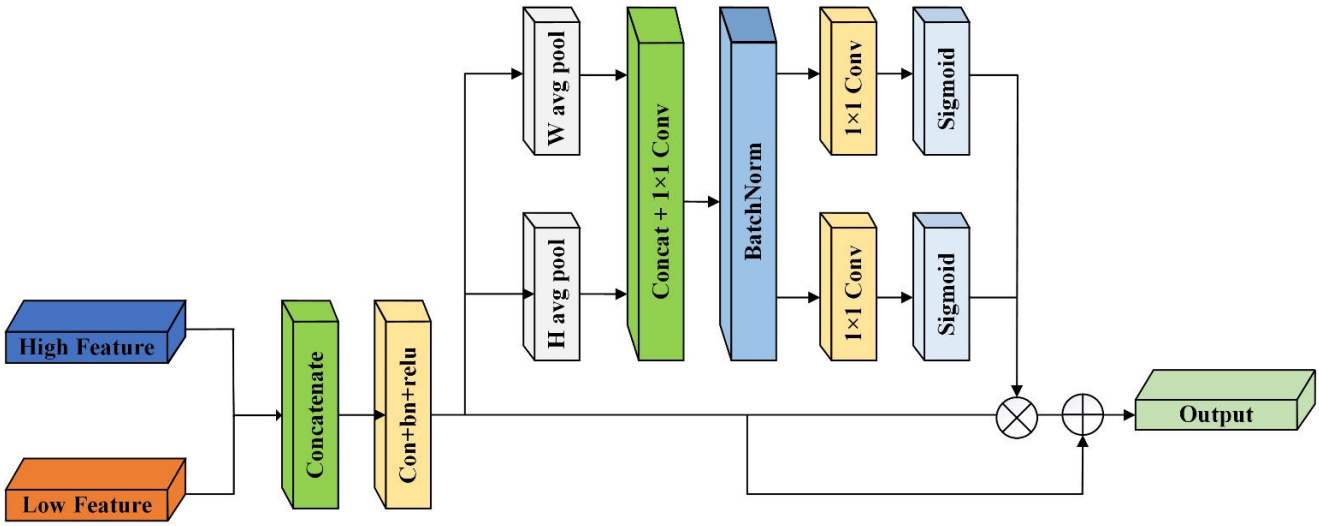



**FIGURE 4.** Refinement extraction module. Combining high-level features and low-level features allows the model to better integrate global and local information for more accurate distress extraction.

### B. REFINEMENT EXTRACTION MODULE

In the step by step feature extraction process of a deep learning network, the network may overemphasize the features of a specific distress, leading to an increased sensitivity to pixel variations of complex distress, which makes the network may produce overly coarse feature extraction when dealing with complex distress such as crack, and it is difficult to capture subtle variations and details. To tackle this issue, the paper introduces the refinement extraction module, which maximizes the utilization of both higher order and lower order features to capture horizontal and vertical pavement distress features. This approach enhances the model's effectiveness in extracting detailed pavement distress information. The refined extraction module is shown in Figure 4.

The refinement extraction module initially overlays the low order and high order features to obtain the feature $F$. The feature sizes are then equalized through the application of batch normalization, yielding the feature $F_c$. Since the morphology of pavement distress mainly consists of transverse and longitudinal directions, we use average pooling in both X and Y axes to split the picture's characteristics into transverse and longitudinal features, which more easily captures the features of pavement distress in these two directions. At H and W, the c-th channel's output can be expressed as:

$$z_c^h(h) = \frac{1}{W} \sum_{0 \le i < W} x_c(h, i)$$
$$z_c^w(w) = \frac{1}{H} \sum_{0 \le j < H} x_c(j, w) \tag{5}$$

Next, the horizontal and vertical feature maps are fused to allow the model to more fully comprehend and utilize the distress features in different directions, and then the convolution is used to learn the weights and interrelationships between the channels by linearly combining the features on the different channels to capture the correlations between the different channels. The expression is as follows:

$$f = \delta(Conv_{1\times1}([\mathbf{z}^h, \mathbf{z}^w])) \tag{6}$$

where $\delta$ is a nonlinear activation function. Afterwards, the feature map is divided into two tensors $\mathrm{f}^h$ and $\mathrm{f}^w$ again along the spatial dimension, and a bidirectional attention vector is generated using the sigmoid function. The attention vector is multiplied with the original feature map to amplify the representation of pavement distress pairs, and the output feature map is subsequently summed with the original features to

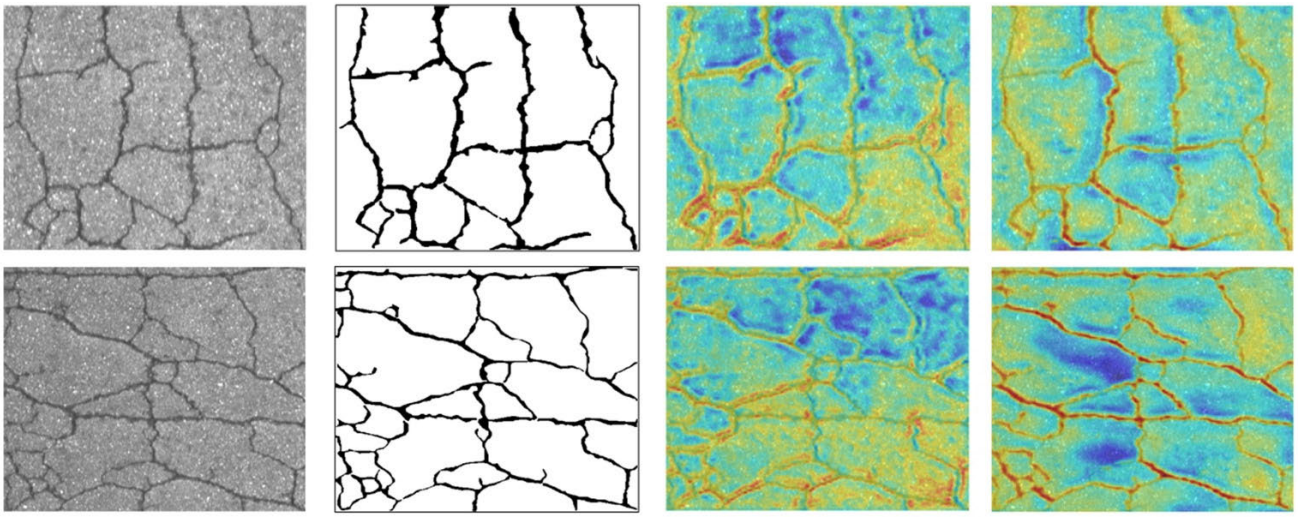



**FIGURE 5.** Feature heatmap visualization during training with or without REM addition. With the use of the REM module the model pay more attention to key information features such as the position, edge of distress in a more precise manner.

prevent degradation, which can locate the edge information of the pavement distresses by using a mixture of high-level and low-level feature maps. The expression is as follows:

$$g^h = \sigma(F_h(f^h))$$
$$g^w = \sigma(F_w(f^w)) \tag{7}$$

The learned features of each channel are activated using the sigmoid function, the weights obtained according to the previous normalization are assigned to the features of each channel. To introduce the attention mechanism into the channel dimension, each channel is multiplied by its corresponding channel coefficient. Figure 5 visually show cases the effectiveness of the distress refinement extraction module with feature heatmaps, illustrating the model's improved ability to capture distress in finer detail.

### C. WEIGHTED CROSS ENTROPY LOSS FUNCTION

When dealing with images of distress, traditional cross entropy loss functions may treat all pixels equally, which can be problematic when the majority of the image represents the background and the distress area is relatively small. This could result in a situation where the contribution of distress regions is relatively minor. In order to address this imbalance issue, this paper proposes a weighted cross entropy loss function. By considering the distance of each pixel in the image to the boundary of the distress, the proposed loss function can finely adjust the weights between foreground (distress region) and background. This allows the model to pay more attention to the distress regions during the training process. The distance information to the distress boundary serves as a weighting factor, enabling a more nuanced adjustment of the contributions from different regions. This approach helps mitigate the imbalance problem, ensuring that the model effectively focuses on and learns from the distress areas

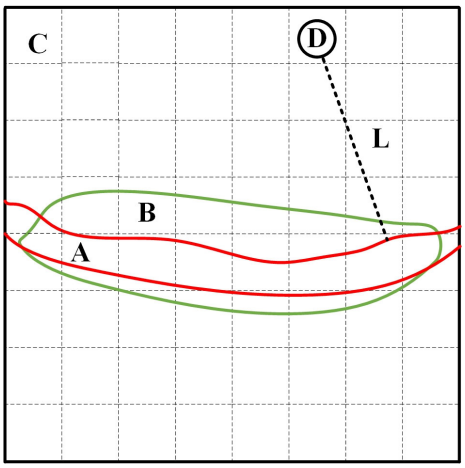



**FIGURE 6.** Schematic diagram of weighted cross entropy loss function. Where A is the real distress area, B is the model prediction, C is the background, D is a pixel point in the image and L is the distance from the point to the true value.

during training. Figure 6 shown a schematic diagram of the weighted cross entropy loss function.

The operation formula is as follows:

$$L_{wec} = -W_{ij}[y_{ij}\log p_{ij} + (1 - y_{ij})\log(1 - p_{ij})] \tag{8}$$

where $W_{ij}$ is the weight at pixel $(i, j)$, $y$ is the label value, and $p$ is the prediction of the model. The operation of is as follows:

1) Find the boundary pixels of $mask_m$. There may be more than one mask in an image, and the boundary pixels may be found according to each independent connected domain.

2) Based on the boundary pixels, a distance transformation operation is performed to obtain the closest distance of each pixel position from the boundary L. Assuming that the input binary image is $I_{(x,y)}$, for each foreground pixel (distressed area), the distance from pixel (x,y) is calculated, and then the

coordinate with the smallest distance is selected as the closest edge pixel's coordinates (i,j). The distance transform $L_{(x,y)}$ can be calculated by the following equation:

$$L_{(x,y)} = \min_{(i,j)} \sqrt{(x-i)^2 + (y-j)^2} \tag{9}$$

where $\min_{(i,j)}$ is the coordinate of the nearest boundary pixel and $\sqrt{(x-i)^2 + (y-j)^2}$ denotes the distance from pixel (x,y) to boundary pixel (i,j). In Figure 6, target D is far from the boundary of the distress; if we directly use the distance as a weight, then it will produce a large loss value, resulting in oscillation in the network optimization process; therefore, we take the logarithm of the distance, and the final $W_{ij}$ is:

$$W_{ij} = \log_2(L+2) \tag{10}$$

## IV. EXPERIMENTS AND RESULTS

### A. TRAINING STRATEGY

The model training model training parameters are set as shown in Table 1. The server used for model training in this study is an Intel(R) Core (TM) i7-9700 CPU and a GPU (Nvidia GeForce RTX 3090), and all the models are implemented using the PyTorch framework.

**TABLE 1. Hyperparameter settings.**

| Parameter | Parameter settings |
|---|---|
| Initial learning rate | 5e-4 |
| Batch size | 4 |
| Optimizer | Adamw |
| lr_decay_type | cos |
| Momentum | 0.95 |
| Epoch | 150 |
| Weight_decay | 1e-4 |

### B. DATA SETS

Five pavement distress datasets were used in this study, Table 2 presents detailed information for 5 datasets.

Most of the existing pavement distress datasets involve cracks, and the number of scenes is low. To address the above problems, this paper uses a vehicle-mounted measurement system to collect pavement images from several highways in Liaoning Province and crops and sketches the pavement distress parts in the collected images to produce a set of pavement distress datasets for the national and provincial arterial highways LNTU_RDD_GS. It contains 300 negative samples and 3149 distress images. The dataset scenes contain eight scenes, and the categories include five categories: transverse crack, longitudinal crack, alligator crack, potholes and repair area; some of the dataset scene categories are shown in Figure 7.

Five pavement distress datasets, including one self-constructed dataset and four publicly available datasets, were used in this study as follows:

1) LNTU_RDD_GS: This dataset was produced by the Spatio Temporal Big Data Research Center of Liaoning University of Engineering and Technology. Pavement images from several highways in Liaoning Province were collected using a vehicle-mounted measurement system, and the broken parts of the collected images were cropped and labelled for outlining.

2) Crack500 [19]: In this dataset, the authors employed a cell phone to capture 500 images of pavement cracks, each approximately 2000 × 1500 pixels in size, with pixel level annotations for each image.

3) CFD [11]: This dataset was compiled from 118 images captured by a cell phone. Each image includes a manually labeled real ground contour, and the captured images feature noise elements like water marks, oil stains, and shadows.

**TABLE 2. Detailed information on datasets.**

| Dataset | Number | Resolution | Annotation level |
|---|---|---|---|
| Crack500 | 500 | 1568×2594 | Patch Level |
| CFD | 118 | 320×480 | Patch Level |
| GAPs384 | 384 | 551×540 | Patch Level |
| Aigle RN | 38 | 522×592 | Patch Level |
| LNTU_RDD_GS | 3449 | 377×373 | Patch Level |

4) GAPs384 [57]: This dataset was constructed by selecting 384 crack images from the GAPs [58] dataset for pixel level annotation to construct a new crack dataset, GAPs384.

5) Aigle RN: The Aigle RN is a collection of 38 images with pixel level annotations obtained from regular monitoring of French road conditions using the Aigle RN system at driving speeds.

### C. EXPERIMENTAL STRATEGY

Two groups of experiments were designed for this experiment. In Experiment 1, the extraction accuracy of DARNet was compared with that of other mainstream models for pavement distress; Experiment 2 was the analysis of the effectiveness of the DARNet method; and the training parameters of all the experiments were the same. In this study, the precision, recall, F score are used as indicators of distress recognition accuracy. mIoU measures the accuracy of the model by calculating the proportion of intersections and concatenations between the segmentation results predicted by the model and the true labels, which also serves as the accuracy metric for this paper. The four indicators are calculated as follows.

$$\begin{aligned} Precision &= \frac{TP}{TP+FP} \\ Recall &= \frac{TP}{TP+FN} \\ F &= \frac{2 \times Precision \times Recall}{Precision + Recall} \\ mIoU &= \frac{1}{2}\left(\frac{TP}{TP+FP+FN} + \frac{TN}{TN+FN+FP}\right) \end{aligned} \tag{11}$$

where $TP$ represents the number of correctly classified pavement distress, $FP$ represents the number of nonpavement distress samples incorrectly classified as pavement distress, and $FN$ represents the number of incorrectly classified nonpavement distress samples.

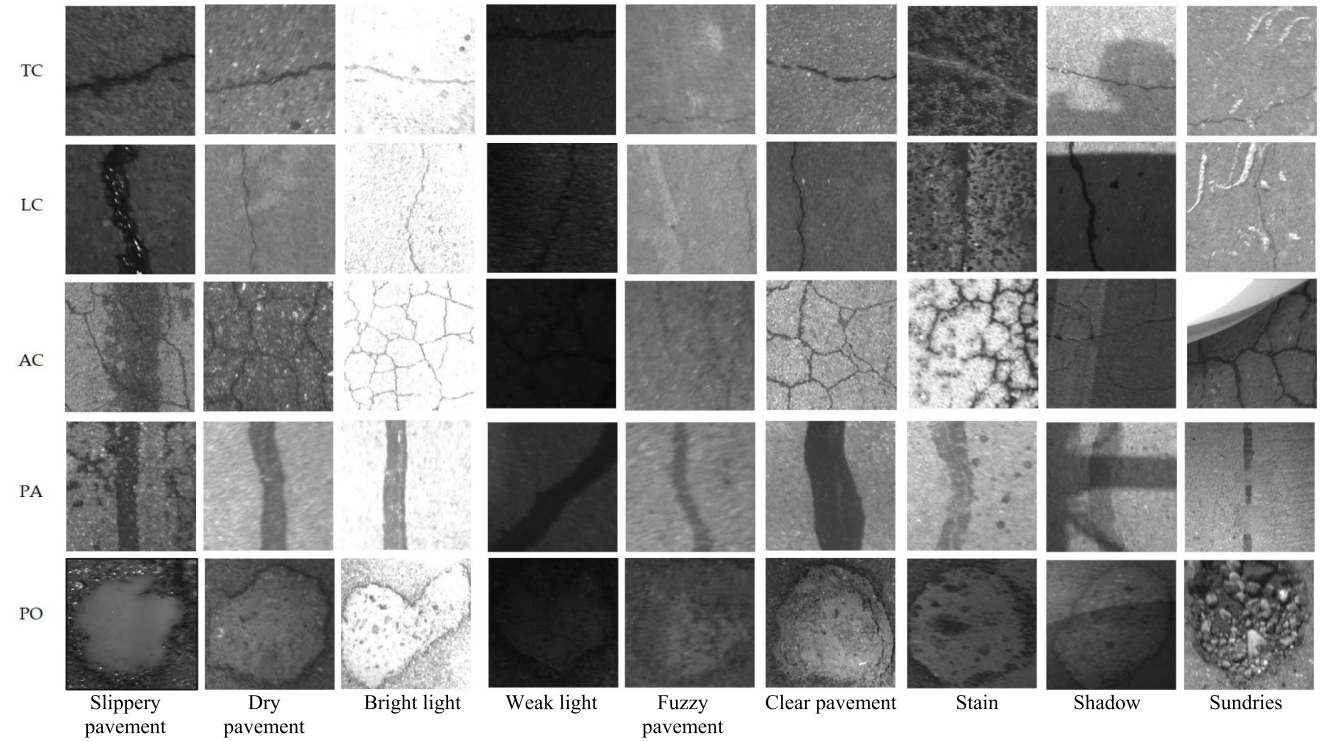



**FIGURE 7.** LNTU_RDD_GS. Where "TC", "LC", "AC", "PA", "PO" are the abbreviations for "transverse crack", "longitudinal crack", "alligator crack", "repair area", "pothole".

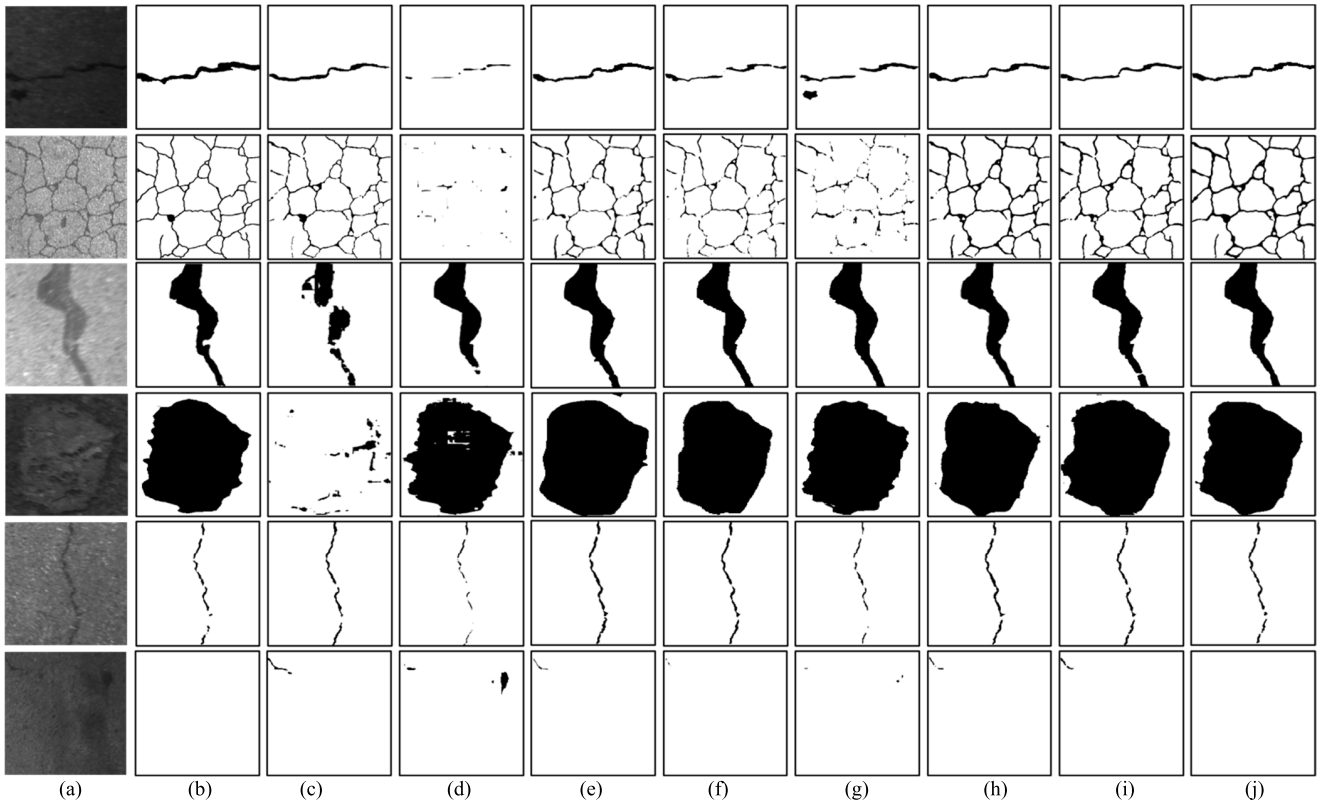



**FIGURE 8.** Results of each model identifying LNTU_RDD_GS dataset. Where (a) Image,(b) GT, (c) UNet,(d) PSPNet, (e) DeepLabv3+, (f) HRNet, (g) SegForme, (h) DeepCrack, (i) FFEDN, (j) DARNet.

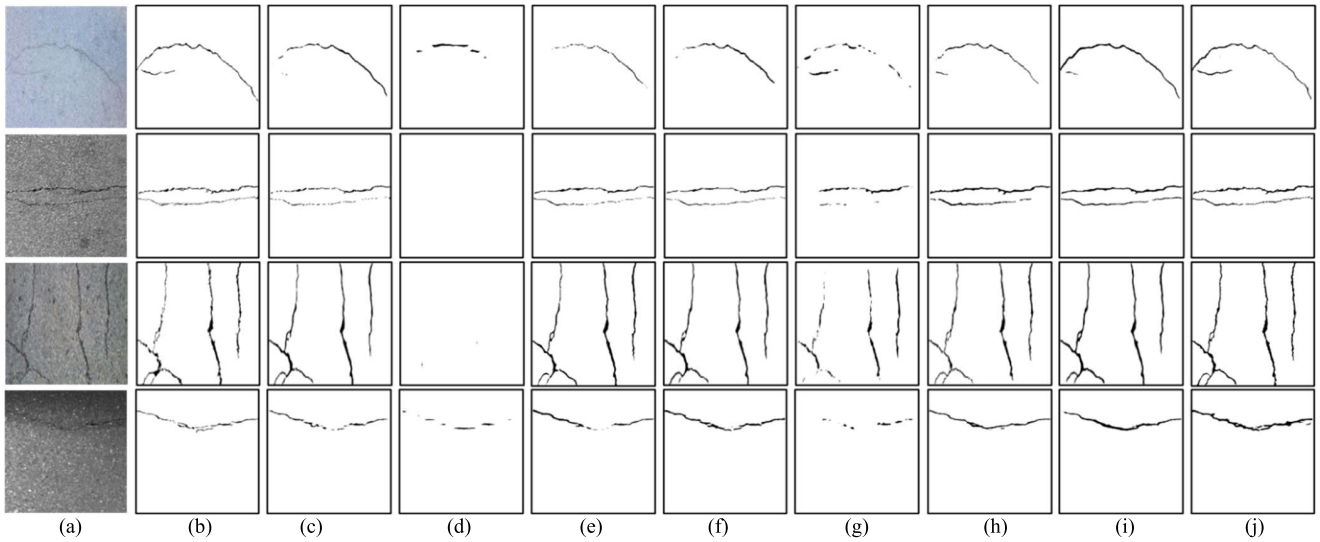



**FIGURE 9.** Results of each model identifying public dataset. Where (a)Image, (b)GT, (c)UNet, (d)PSPNet, (e)DeepLabv3+, (f)HRNet, (g)SegFormer, (h),DeepCrack, (i)FFEDN, (j)DARNet.

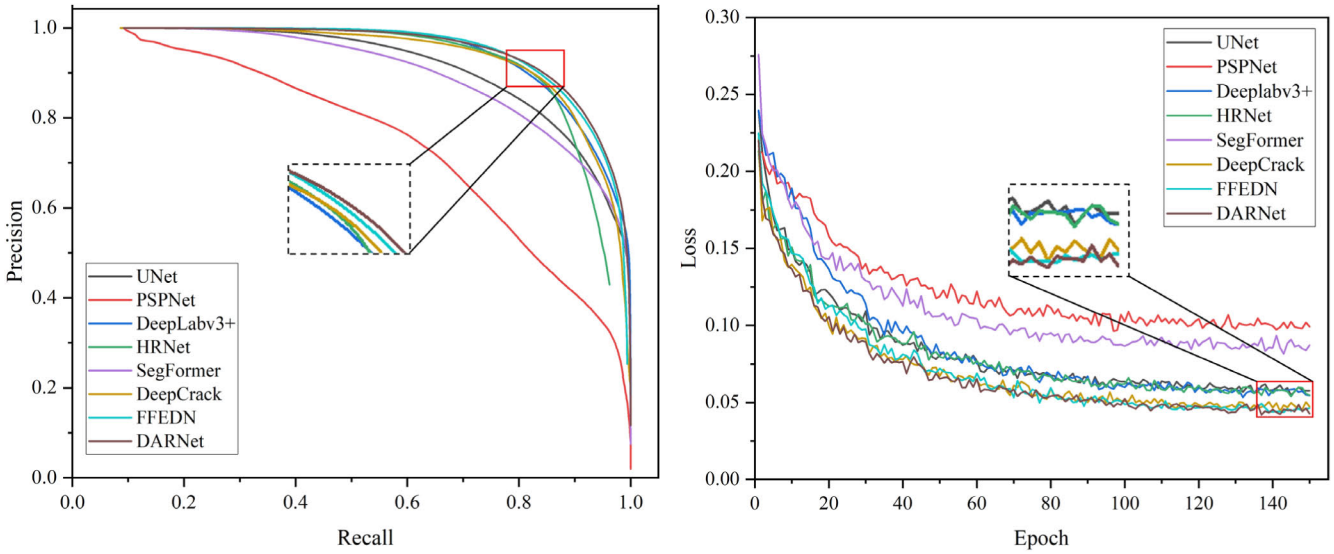



**FIGURE 10.** PR and Loss curves for each model on the LNTU_RDD_GS.

**TABLE 3.** Model identification results for the LNTU_RDD_GS dataset.

| Metric | UNet | PSPNet | DeepLabv3+ | HRNet | SegFormer | DeepCrack | FFEDN | DARNet |
|---|---|---|---|---|---|---|---|---|
| mIoU | 77.16 | 61.70 | 80.88 | 78.29 | 71.71 | 81.24 | 81.47 | 82.68 |
| Recall | 82.43 | 67.32 | 86.32 | 83.06 | 79.68 | 87.54 | 87.81 | 88.13 |
| Precision | 90.05 | 76.71 | 90.53 | 91.27 | 83.09 | 90.12 | 90.23 | 92.22 |
| F score | 86.07 | 71.71 | 88.37 | 86.97 | 81.35 | 88.81 | 89.00 | 90.13 |

### *D. MODEL COMPARISON EXPERIMENT*

To compare the performance of DARNet with that of other mainstream networks, UNet [59], DeepLabv3+ [60], PSPNet [61], HRNet [62], the Segformer [63], DeepCrack [24], and FFEDN [64] are trained with the LNTU_RDD_GS dataset.

Figure 8 illustrates the results of each model recognition test set. PSPNet performs relatively poorly at all scales, especially at the larger scale of pothole or repair area, which is al-most impossible for the model to recognize correctly. SegFormer is less effective at identifying complex alligator crack and can misidentify some stains, shadows and other disturbances, as distress. UNet is less effective at identifying potholes and repair area. Its missed recognition rate is relatively high, which is especially obvious in its recognition of Cracks. The overall recognition effect of HRNet is better than that of the other methods, but missed recognition due to tiny cracks still occurs. DeepCrack and FFEDN have

**TABLE 4.** Model identification results for the publicly datasets.

| Metric | UNet | PSPNet | DeepLabv3+ | HRNet | SegFormer | DeepCrack | FFEDN | DARNet |
|---|---|---|---|---|---|---|---|---|
| mIoU | 79.72 | 66.59 | 78.46 | 79.52 | 70.95 | 80.05 | 80.35 | 80.63 |
| Recall | 84.84 | 70.66 | 82.13 | 84.58 | 73.91 | 88.20 | 85.43 | 84.84 |
| Precision | 90.52 | 83.72 | 92.24 | 90.50 | 90.41 | 87.25 | 90.83 | 92.17 |
| F score | 87.59 | 76.64 | 86.89 | 87.43 | 81.33 | 87.72 | 88.05 | 88.35 |

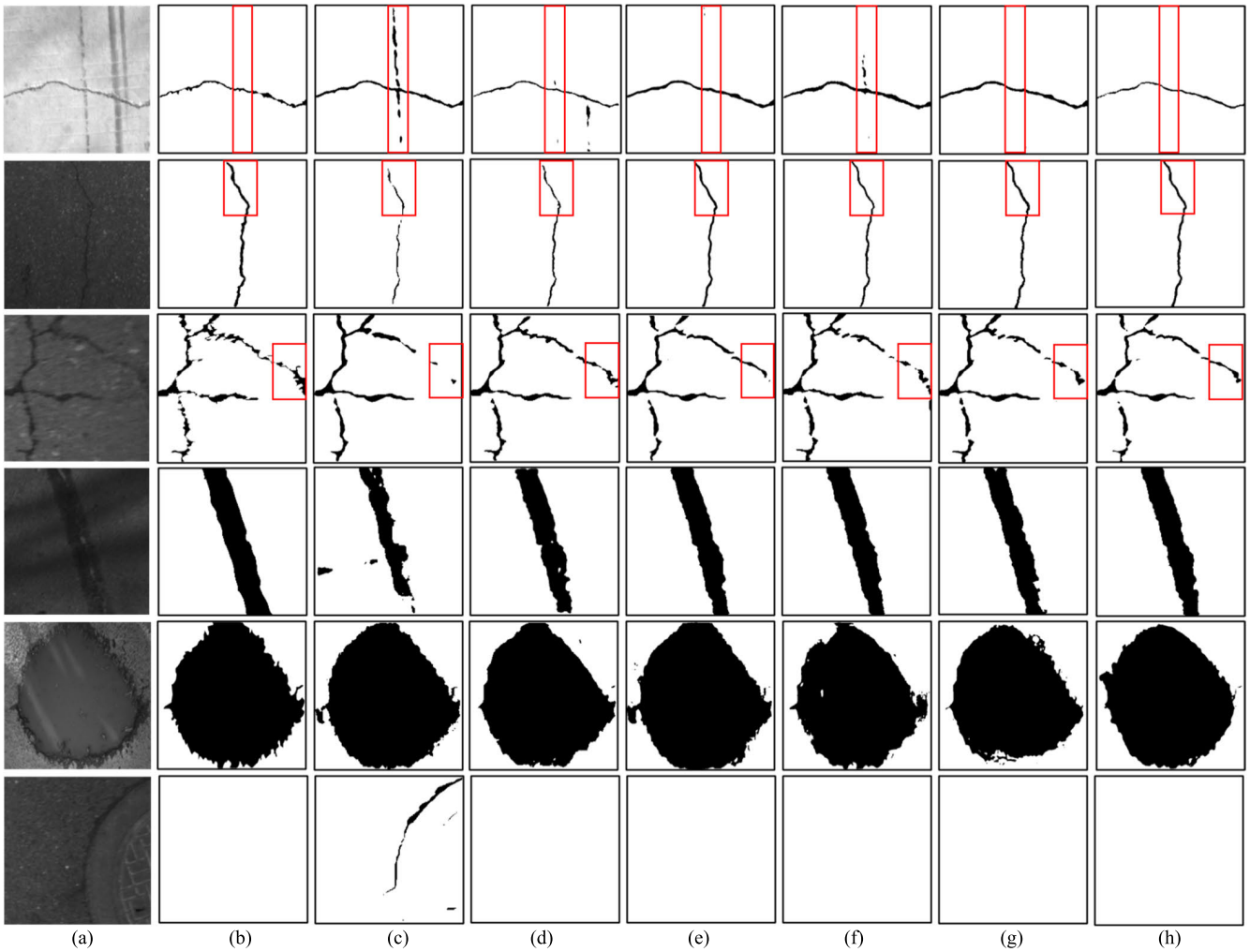



**FIGURE 11.** Added different models to identify the LNTU_RDD_GS results. Where (a)Image, (b)Ground Truth, (c)DeepLabv3+, (d)CFASPP, (e)DAAM, (f)REM, (g)WCEL, (h)DARNet.

better overall recognition results but are prone to breakage when recognizing cracks. In contrast, the proposed method performs more finely and continuously than the other models when recognizing cracks and can maintain good results when recognizing high-resolution images. For distress like pothole and repair area, the proposed method can also capture additional details and better recognize lesions in harsh environments. Figure 10 shows the loss and PR curves for each model training.

Four previously mentioned public datasets are employed in this research to confirm the universality of the model technique, and due to the small number of datasets, they are combined for training; the same method is used in [65], and the training results are shown in Figure 9. Due to the different labelling methods and pavement materials in the four datasets, the recognition effects are different. The cracks in the Crack500 dataset are the clearest, and all the model recognition effects are more complete except for PSPNet and SegFormer. In contrast, in CFD, AEL, and GAPs384, there are many tiny cracks in the image, and the image is impacted by uneven illumination and noise interference; thus, the models tend to miss identifying distress. The ability of DARNet to extract the details of the distress is significantly superior to that of the other comparison methods.

Table 3 shows the quantitative results of each model on the LNTU_RDD_GS dataset. DARNet obtained optimal results with mIoU, recall, precision and F score values of 82.68%, 88.13%, 92.22% and 90.13%, respectively. Compared with

**TABLE 5.** The results of different backbone on the accuracy of DARNet.

| Backbone | mIoU | Recall | Precision | F score | Parameter |
|---|---|---|---|---|---|
| VGG19 | 81.93 | 85.57 | 91.38 | 88.38 | 143,667,240 |
| DenseNet201 | 82.01 | 86.28 | 91.44 | 88.79 | 20,242,984 |
| ResNet152 | 82.15 | 86.99 | 91.63 | 89.25 | 60,192,808 |
| InceptionV3 | 82.37 | 87.49 | 91.85 | 89.62 | 23,626,728 |
| Xception | 82.68 | 88.13 | 92.22 | 90.13 | 22,855,952 |

**TABLE 6.** Results of ablation experiments.

| Baseline | CFASPP | DAAM | REM | WCEL | mIoU | F score |
|---|---|---|---|---|---|---|
| √ | | | | | 80.88 | 88.37 |
| | √ | | | | 81.32 | 89.72 |
| | | √ | | | 81.67 | 89.67 |
| | | | √ | | 82.01 | 90.02 |
| | | | | √ | 81.52 | 89.95 |
| | √ | √ | √ | √ | 82.68 | 90.13 |

those of DeepLabv3+, the mIoU and recall improved by 1.80%, and the F score improved by 1.76%.

Table 4 shows the quantitative results of each model on publicly available datasets. DARNet obtained optimal results with mIoU, recall, precision and F score values of 80.63%, 84.84%, 92.17% and 88.35%, respectively. Compared with those of DeepLabv3+, the mIoU and recall improved by 2.17%, and the F score improved by 1.46%.

### E. ABLATION EXPERIMENT

In order to compare the impact of different backbone networks on DARNet, this article uses different networks as the backbone of DARNet, and the experimental results are shown in Table 5. Xception achieved the best accuracy.

In order to confirm the efficacy of the model, four sets of ablation experiments are conducted to evaluate the effects of CFASPP, DAAM, REM and WCEL on the model. The results in Table 6 indicate that each module contributes to enhancing segmentation accuracy. This demonstrates the effectiveness of each module

The results of adding various modules to the recognition of the LNTU_RDD_GS test set are displayed in Figure 11. In the first row, each module reduces the misrecognition of this stain. In the third row, each module addresses the missed recognition, and in the last row, each module avoids misrecognition. CFASPP fully utilizes the features between different scales to extract distresses, and the recognition of each distress is improved. With the inclusion of the DAAM module to help the network focus on the distressed area during upsampling, object under recognition is significantly reduced. The addition of the REM module fully utilizes the higher order features and lower order feature information to enable finer distress extraction; each module effectively enhances the feature capability of the network.

## V. CONCLUSION

This work aims to solo the problem of rough pavement distress extraction and improve the accuracy of tiny pavement distress identification. Therefore an image segmentation algorithm for pavement distress in DARNet is proposed. CFASPP utilizes different scales of convolution to extract features and realizes the reuse of features in each layer, which improves the model's performance of localization and segmentation of pavement distresses in different scales, and simultaneous with, it can capture the shape and structural information of pavement distresses in a better way. The DAAM module utilizes channel and spatial attention mechanisms to focus the model on areas related to key channels and distress patterns, enhancing the perception of spatial structure and reducing the blurring or offset problems introduced by upsampling. The REM combines high- and low-order features to capture transverse and longitudinal pavement distresses, which significantly improves the extraction of pavement distress information details. The weighted cross entropy loss function to assign weights based on the distance from pixels to distress of boundary. Comparison with other mainstream models using the model in this paper, and the results show that DARNet can achieve an 82.68% mIoU and an 90.13% F score on the LNTU_RDD_GS test set, and an 80.63% mIoU and 88.35% F score in the four public datasets.

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